FinGuard — Technical Paper

Abstract

FinGuard is a portfolio simulation and risk management tool that applies the **Kelly Criterion** for optimal position sizing, integrates **drawdown controls**, and evaluates portfolio robustness using **Monte Carlo simulations**. Designed as an educational project, FinGuard demonstrates the application of quantitative finance techniques in a reproducible, test-driven framework. This paper describes the system architecture, algorithms, evaluation methodology, and future development.

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1. Introduction

Managing risk is central to investment strategy. While traditional portfolio optimization focuses on mean-variance tradeoffs, FinGuard implements more dynamic risk tools: Kelly Criterion for growth maximization, drawdown constraints for capital preservation, and Monte Carlo simulations for uncertainty modeling. The system enables users to visualize risk/reward tradeoffs and assess strategies under different market conditions.

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2. System Architecture

FinGuard is structured as a modular pipeline:

1. **Portfolio Definition**: User defines assets and allocations.
2. **Kelly Sizing**: Optimal allocation weights computed using Kelly Criterion.
3. **Drawdown Control**: Enforces capital allocation adjustments when losses exceed thresholds.
4. **Monte Carlo Simulation**: Generates synthetic return paths to stress test portfolios.
5. **Visualization**: Streamlit dashboard with plots of equity curves, histograms, and drawdown charts.

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3. Methods and Algorithms

3.1 Kelly Criterion

- Formula:

$$f^* = \frac{bp - q}{b}$$

Where:
- f^* : optimal fraction of capital to invest
- b : odds received per unit bet (risk/reward ratio)
- p : probability of winning

- $q = 1 - p$: probability of losing

- In portfolio context, Kelly weights are derived from asset expected returns and covariance matrices.

3.2 Drawdown Control

- Tracks running peak portfolio value.
- If current value drops below peak by a threshold (e.g., 20%), allocation is reduced or strategy halts.
- Protects against ruin scenarios during adverse sequences.

3.3 Monte Carlo Simulation

- Randomized sampling from historical or synthetic return distributions.
- Simulates thousands of portfolio paths to estimate distribution of outcomes.
- Key metrics: Expected CAGR, max drawdown, Value-at-Risk (VaR).

3.4 Visualization

- **Equity Curve**: Portfolio value trajectory.
- **Histogram**: Return distribution from simulations.
- **Drawdown Plot**: Depth and duration of portfolio losses.

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4. Implementation

- **Language**: Python
- **Libraries**: NumPy, Pandas, matplotlib/Plotly, Streamlit.
- **Testing**: Pytest with unit, integration, and reproducibility tests.
- **Design Pattern**: Separation of pure financial logic (in `kelly.py`, `drawdown.py`, `simulator.py`) from visualization/UI (`visualizer.py`, `app.py`).

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5. Evaluation

5.1 Dataset

- Historical returns for assets such as BTC, ETH, SPY.
- Synthetic Gaussian and bootstrapped return paths for Monte Carlo.

5.2 Metrics

- **Expected CAGR**: compound annual growth rate.
- **Max Drawdown**: worst peak-to-trough loss.
- **Sharpe Ratio**: return/risk tradeoff.
- **VaR (95%)**: Value-at-Risk at 95% confidence.

5.3 Results (Pilot)

- Kelly allocation outperforms equal weighting on growth but increases drawdown risk.
- Monte Carlo stress tests highlight variance in crypto-heavy portfolios.
- Drawdown control reduces tail risk but lowers expected growth.

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6. Testing Strategy

- **Unit Tests**: Kelly formula correctness, drawdown calculations.
- **Integration Tests**: Portfolio simulation end-to-end.
- **Monte Carlo Reproducibility**: Fixed seed tests to verify distribution consistency.
- **Visualization Integrity**: Validate plot data vs. simulation results.

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7. Discussion

FinGuard balances growth and risk in a simulated environment. The Kelly Criterion offers mathematically optimal allocations but can be overly aggressive without drawdown safeguards. Monte Carlo analysis reveals risk exposure under uncertain future conditions. Together, these methods provide a deeper understanding of portfolio dynamics than static optimization approaches.

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8. Future Work

- Add Value-at-Risk and Conditional VaR modules.
- Integrate real-time data feeds (e.g., FRED, crypto APIs).
- Implement rebalancing and transaction cost models.
- Compare with alternative optimization methods (mean-variance, Black-Litterman).

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9. Conclusion

FinGuard demonstrates how mathematical risk models can be combined into an interactive, test-driven simulation tool. While not intended for production trading, it provides a reproducible framework for exploring portfolio optimization and risk management concepts.

References

- Kelly, J. L. (1956). *A New Interpretation of Information Rate*. Bell System Technical Journal.
- Monte Carlo Methods in Finance: Glasserman, P. (2004).
- Risk Management and Financial Institutions: Hull, J. (2018).
- NumPy: https://numpy.org
- Streamlit: https://streamlit.io